TESLA COACHING CLASSES

Mathematics – Part 1 (Algebra)

Time: 2 Hours	Standard: Class 10	Total Marks: 40

Part Syllabus Test - 08 (PST-08)

Student Name: _____ **Date:** _____

CHAPTERS COVERED & WEIGHTAGE

CHAPTER 1	Linear Equations in Two Variables	30%
CHAPTER 2	Quadratic Equations	30%
CHAPTER 3	Arithmetic Progression	20%
CHAPTER 4	Financial Planning	20%

Q.1(A) **Choose the correct alternative from given:** **[4 Marks]**

i) To solve $x + y = 3\,;\, 3x - 2y - 4 = 0$ by determinant method find D.

(a) 5 **(b)** 1

(c) -5 **(d)** -1

ii) Out of the following equations, find the equation having the sum of its roots -5.

(a) $3x^2 - 15x + 3 = 0$ **(b)** $x^2 - 5x + 3 = 0$

(c) $x^2 + 3x - 5 = 0$ **(d)** $3x^2 + 15x + 3 = 0$

iii) For an given A.P. $t_7 = 4$, $d = -4$ then $a = \ldots$

(a) 6 **(b)** 7

(c) 20 **(d)** 28

iv) If the Face Value of a share is ₹ 100 and Market value is ₹ 75, then which of the following statements is correct ?

(a) The share is at premium of ₹ 175 **(b)** The share is at discount of ₹ 25

(c) The share is at premium of ₹ 25 **(d)** The share is at discount of ₹ 75

Q.1(B) **Solve the following subquestions:** **[4 Marks]**

i) Write any two solutions of the equation $x + y = 7$.

ii) Write the given quadratic equation in standard form and also write the values of a, b and c.
$4y^2 - 3y = -7$

iii) Write the formula of the sum of first n terms for an A.P.

iv) 'Pawan Medical' supplies medicines. On some medicines the rate of GST is 12%, then what is the rate of CGST and SGST?

Q.2(A) **Complete and write any TWO activities from the following:** **[4 Marks]**

i) If $(2, 0)$ is the solution of $2x + 3y = k$, then find the value of k by completing the activity.

$(2, 0)$ is solution of equation $2x + 3y = k$.

Putting $x =$ ☐ and $y =$ ☐

$\therefore 2($ ☐ $) + 3 \times 0 = k$

$\therefore 4 + 0 = k$

$\therefore k =$ ☐

ii) Solve : $x^2 + 8x - 48 = 0$ by factorisation. Complete the following activity.

$x^2 + 8x - 48 = 0$

$\therefore x^2 +$ ☐ $-4x - 48 = 0$

$\therefore x(x + 12) - 4($ ☐ $) = 0$

$\therefore (x + 12)($ ☐ $) = 0$

$\therefore x + 12 = 0$ or $x - 4 = 0$

$\therefore x = -12$ or $x =$ ☐

iii) Which term of following A.P. is -940. $50, 40, 30, 20 \ldots$

Here $a =$ ☐ , $d =$ ☐ , $t_n = -940$

According to formula, $t_n = a + (n - 1)d$

$-940 =$ ☐

$n =$ ☐

Q.2(B) **Solve any FOUR subquestions from the following:** **[8 Marks]**

i) If $(2, -5)$ is the solution of the equation $2x - ky = 14$, then find k.

ii) Find the values of following determinants. $\begin{vmatrix} -1 & 7 \\ 2 & 4 \end{vmatrix}$

iii) One of the roots of quadratic equation $2x^2 + kx - 2 = 0$ is -2, find k.

iv) $11, 8, 5, 2, \ldots$ In this A.P. which term is number -151?

v) The taxable value of a wrist watch belt is ₹ 586. Rate of GST is 18%. Then what is price of the belt for the customer ?

Q.3(A)	Complete and write any ONE activity from the following:	[3 Marks]

i) A two digit number and the number with digits interchanged add up to 143. In the given number the digit in unit's place is 3 more than the digit in the ten's place. Find the original number.

Let the digit in unit's place is x and that in the ten's place is y

$\therefore$ the number $=$ $\square$ $y + x$

The number obtained by interchanging the digits is $\square$ $x + y$

According to first condition two digit number $+$ the number obtained by interchanging the digits $= 143$

$\therefore 10y + x + 10x + y = 143$

$\therefore 11x + 11y = 143$

$x + y = 13 \quad \ldots\ldots (\mathrm{I})$

From the second condition, digit in unit's place $=$ digit in the ten's place $+3$

$\therefore x = y + 3$

$\therefore x - y = 3 \quad \ldots\ldots (\mathrm{II})$

Adding equations (I) and (II)

$2x =$ $\square$

$x = 8$

Putting this value of x in equation (I)

$x + y = 13$

$8+$ $\square$ $= 13$

$\therefore y =$ $\square$

The original number is $10y + x$

$=$ $\square$ $+8$

$= 58$

ii) Complete the following activity to solve the given word problem. Sum of squares of two consecutive even natural numbers is 244, then find those numbers.

Let the first even natural number be x .

Therefore its consecutive even natural number will be $=$ $\square$

By the given condition,

$x^2 + (x + 2)^2 = 244$

$x^2 + x^2 + 4x + 4-$ $\square$ $= 0$

$2x^2 + 4x - 240 = 0$

$x^2 + 2x - 120 = 0$

x^2+ $\square$ $-$ $\square$ $-120 = 0$

$x(x + 12)-$ $\square$ $(x + 12) = 0$

$(x + 12)(x - 10) = 0$

$x =$ $\square$ / $x = 10$

But natural number cannot be negative, $x = -12$ is not possible.

Therefore first even natural number is $x = 10$.

Second even consecutive natural number $= x + 2 = 10 + 2 = 12$.

Q.3(B)	Solve any TWO subquestions from the following:	[6 Marks]

i) Solve the following simultaneous equations using Cramer's rule. $x + 2y = -1$; $2x - 3y = 12$

ii) Solve the following quadratic equation by the formula method: $5m^2 - 4m - 2 = 0$

iii) For an A.P., $t_4 = 12$ and its common difference $d = -10$, then find t_n.

iv) Prashant bought 50 shares of FV ₹ 100, having MV ₹ 180. Company gave 40% dividend on the shares. Find the rate of return on investment.

Q.4	**Solve any TWO subquestions from the following:**	**[8 Marks]**

i) Sum of the present ages of Manish and Savita is 31. Manish's age 3 years ago was 4 times the age of Savita. Find their present ages.

ii) Suyash scored 10 marks more in second test than that in the first. 5 times the score of the second test is the same as square of the score in the first test. Find his score in the first test.

iii) Shubhankar invested in a national savings certificate scheme. In the first year he invested ₹ 500, in the second year ₹ 700, in the third year ₹ 900 and so on. Find the total amount that he invested in 12 years.

Q.5	**Solve any ONE subquestion from the following:**	**[3 Marks]**

i) Ranjana wants to distribute 540 oranges among some students. If 30 students were more each would get 3 oranges less. Find the number of students.

ii) The total value (with GST) of a remote-controlled toy car is ₹ 1770. Rate of GST is 18% on toys. Find the taxable value, CGST and SGST for this toy-car.

ALL THE BEST

DESIGNED BY YOUR MENTOR – LUCKY BHAIYA